

Summary -D4

1. Idempotent Laws
 - a. $p \equiv p \vee p$.
 - b. $p \equiv p \wedge p$.
2. Commutative Laws
 - a. $p \wedge q \equiv q \wedge p$.
 - b. $p \vee q \equiv q \vee p$.
3. Associative Laws
 - a. $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.
 - b. $p \vee (q \vee r) \equiv (p \vee q) \vee r$.
4. Distributive Laws
 - a. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
 - b. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
5. De Morgan's Laws
 - a. $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
 - b. $\neg(p \vee q) \equiv \neg p \wedge \neg q$
6. Conditional identity

$$p \Rightarrow q \equiv \neg p \vee q$$
7. Complement Law

$$\neg(\neg p) \equiv p$$
8. Identity laws
 - a. $p \wedge \mathbf{T} \equiv p$
 - b. $p \vee \mathbf{F} \equiv p$
9. Domination laws
 - a. $p \vee \mathbf{T} \equiv \mathbf{T}$
 - b. $p \wedge \mathbf{F} \equiv \mathbf{F}$
10. Negation laws
 - a. $p \vee \neg p \equiv \mathbf{T}$
 - b. $p \wedge \neg p \equiv \mathbf{F}$

Ex.

Using logically equivalent statements, without the direct use of truth tables,

1. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent
2. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.
3. Show that $(p \wedge q) \Rightarrow (p \vee q)$ is a tautology.
4. Show that $\sim(\sim p \wedge q) \wedge (p \vee q) = p$

Consider the following statements

- For any rational number x , $x^2 \neq 2$.
- There exists an integer n such that $n^2 + 2n = 0$
- For any positive real number x there exists natural number n such that $\frac{1}{n} < x$.
- Given any real numbers a and b if $a < b$, then there exist a rational number r such that $a < r < b$.
- For all real numbers a and b , $|ab| = |a||b|$.

First order logic

We studied propositional logic. We used propositional forms to represent sentences that were either true or false. The sentences of mathematics involve ideas that cannot be fully represented in propositional logic. These sentences are able to characterize objects, such as numbers or geometric figures, by describing properties of the objects, such as being even or being a rectangle, and the relationships between them, such as equality or congruence. Since propositional logic is not well suited to handle these ideas, we extend propositional logic to a richer system.

Statements in First-Order Logic (FOL) are constructed using following components:

- Predicates: $P(x), Q(x, y), \dots$
- Logical Connectives: $\sim, \wedge, \vee, \Rightarrow, \Leftrightarrow$
- Quantifiers: The symbols $\forall$ (**Universal**, "for all") and $\exists$ (**Existential**, "there exists").

Predicates

Statements involving variables, such as

$"x > 3," "x = y + 3," "x + y = z,"$

and

"computer x is functioning properly,"

These statements are neither true nor false when the values of the variables are not specified. In this section, we will discuss the ways that propositions can be produced from such statements.

The statement "***x is greater than 3***" has two parts. The first part, the variable x , is the subject of the statement. The second part the **predicate**, "***is greater than 3***"—refers to a property that the subject of the statement can have. We can denote the statement "***x is greater than 3***" by $P(x)$, where P denotes the predicate "***is greater than 3***" and x is the variable. The statement $P(x)$ is also said to be the value of the propositional function P at x . Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

Note:

Consider the following ^{Sentence} ~~statement~~: $x > 3$.

The truth value of the ~~statement~~ ^{Sentence} has no meaning without specifying the value of x .

However, we *can* make propositions out of such ^{Sentence} ~~statements~~.

A *predicate* is a property that is true or false about the *subject* (in logic, we say "variable") of a *statement*.

$$\underbrace{x}_{\text{subject}} \underbrace{\text{is greater than } 3}_{\text{predicate}}$$

Consider following examples:

01. Let $P(x)$ denote the statement " $x > 3$." What are the truth values of $P(4)$ and $P(2)$?

$P(x): x > 3$

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement " $x > 3$."

Hence, $P(4)$, which is the statement " $4 > 3$," is true. However, $P(2)$, which is the statement " $2 > 3$," is false.

02. Let $Q(x, y)$ denote the statement " $x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

$Q(x, y): x = y + 3$

Solution: To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$.

Hence, $Q(1, 2)$ is the statement " $1 = 2 + 3$," which is false. The statement $Q(3, 0)$ is the proposition " $3 = 0 + 3$," which is true.

Definition: An open statement (also called a **predicate**) is a sentence that contains one or more variables and whose truth value depends on the values assigned for the variables. We represent an open statement by a capital letter followed by the variable(s) in parenthesis, e.g., $P(x)$, $Q(x)$, $R(x, y)$, ...etc

Note:

- A predicate is neither true nor false
- A predicate becomes a proposition when the variables are substituted with specific values
- The **domain** of a predicate variable is the set of all values that may be substituted for the variable

Definition: The domain or universe or universe of discourse for a predicate variable is the set of values that may be assigned to the variable.

Symbol	Predicate	Domain	Truth set
$p(x)$	$x > 3$	$\mathbb{R}$	$\{x x \in \mathbb{R} \text{ and } x > 3\}$
$p(x,y)$	$x + 2y$ is odd	$\{(x,y) x \in \mathbb{Z}, y \in \mathbb{Z}\}$	$\{(x,y) x \in \mathbb{Z}, x \text{ is odd and } y \in \mathbb{Z}\}$
$p(x)$	$x > 1 \Rightarrow x > \frac{1}{\sqrt{3}}$	$\mathbb{R}$	$\mathbb{R}$
$p(x)$	$\sqrt{5-x} > 10$	$\{x x \in \mathbb{R} \text{ and } x \leq 5\}$	$\{x x \in \mathbb{R} \text{ and } x < -95\}$

- **Predicate:** An expression $P(x)$ defined on a given set A is called **Predicate** if $P(x)$ has the property that $P(a)$ is true or false for each a in A .
- i.e. $P(a)$ becomes a statement (with a truth value) whenever any element a in A is substituted for the variable x .

Example for predicate

- (i). $x^2 > 6$ is a predicate involving the variable x , where x is a natural number.

Let $P(x): x^2 > 6$

Then we have, $P(1), P(2)$ are false, but when $x \in \mathbb{N}$ and $x \neq 1, x \neq 2, P(x)$ is true.

- (ii) $x^2 + y^2 = 4$ is a predicate with two variables x and y , where x, y are real numbers.

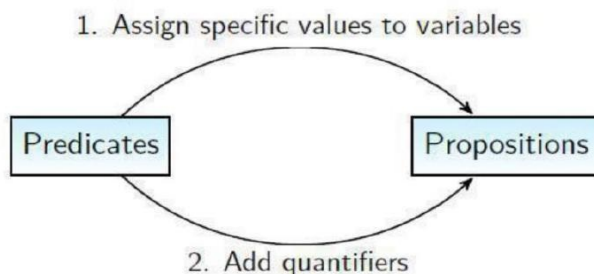
Let $P(x, y): x^2 + y^2 = 4$

Then $P(0,2)$ is true, but $P(1,1)$ is false.

Predicates to propositions

There are two methods to obtain propositions from predicates

1. Assign specific values to variables
2. Add quantifiers



Quantifiers

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value. However, there is another important way, called quantification, to create a proposition from a propositional function. Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words **all**, **some**, **many**, **none**, and **few** are used in quantifications. We will focus on two types of quantification here: **universal quantification**, which tells us that a predicate is true for every element under consideration, and **existential quantification**, which tells us that there is one or more element under consideration for which the predicate is true. The area of logic that deals with predicates and quantifiers is called the predicate calculus.

Note: Quantifiers

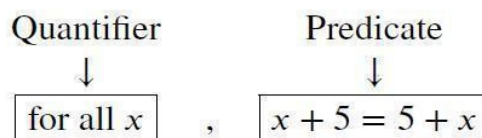
A predicate becomes a proposition when we assign it fixed values. However, another way to make a predicate into a proposition is to *quantify* it. That is, the predicate is true (or false) for *all* possible values in the universe of discourse or for *some* value(s) in the universe of discourse.

Such *quantification* can be done with two *quantifiers*: the *universal* quantifier and the *existential* quantifier.

- **Quantifiers** are words that refer to quantities such as “all” or “some” and they tell for how many elements a given predicate is true
- Two types of quantifiers:
 1. Universal quantifier ($\forall$)
 2. Existential quantifier ($\exists$)

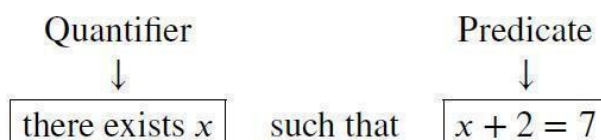
Assume that x is a real number. The sentence

for all x , $x + 5 = 5 + x$



Assume that x represents a real number and consider the sentence

there exists x such that $x + 2 = 7$.



Definition. Let x be a variable whose domain contains the set S , and let $P(x)$ be an open sentence. The *universal quantification* of $P(x)$ over S , written

$$\forall x \in S, P(x),$$

is a statement that is true if $P(x)$ is true for each choice of $x \in S$, and it is false if $P(x)$ is false for at least one choice of $x \in S$.

The symbol $\forall$ is read as any of the following (depending on context): “for all”, “for every”, “for any”, and “for each”. Thus, one may read the statement $\forall x \in S, P(x)$ as “for each x in S , $P(x)$ holds.”

Examples:

01. For every $n \in \mathbb{Z}$, $2n$ is even, can be expressed following in the following way.

$$\forall n \in \mathbb{Z}, 2n \text{ is even}$$

02. Let $\mathbb{R}$ be the universal set. Then, for each real number x , $x^2 > 0$, can be expressed in the following way.

$$\forall x \in \mathbb{R}, x^2 > 0$$

positive

$$\text{Let } x=0: x^2 > 0$$

$$\forall x \in \mathbb{R}, x^2 > 0$$

03. For each integer n , $(n^2 + n + 1)$ is a prime number, can be expressed in the following way.

$$\forall n \in \mathbb{N}, (n^2 + n + 1) \text{ is prime.} \text{ is F } (n=2, n^2+n+1=21=3 \times 7 \text{ not prime})$$

Definition. Let x be a variable whose domain contains the set S , and let $P(x)$ be an open sentence. The *existential quantification* of $P(x)$ over S , written

$$\exists x \in S, P(x),$$

is a statement that is true if $P(x)$ is true for at least one choice of $x \in S$, and it is false if $P(x)$ is false for each choice of $x \in S$.

The statement $\exists x \in S, P(x)$ reads as “there exists some x in S such that $P(x)$.”

Examples:

01. There is an integer n for which $n^2 = 2$.

$$\exists n \in \mathbb{Z}, n^2 = 2. \text{ — F}$$

02. There exists an integer x such that $3x - 2 = 0$.

$$\exists x \in \mathbb{Z}, 3x - 2 = 0 \text{ is a F Statement (Since for each } x \in \mathbb{Z}, 3x - 2 \neq 0)$$

03. For each real number x , there exists a real number y such that $x + y = 0$.

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R}, x + y = 0 \text{ is a T Statement (Since let } x \in \mathbb{R} \text{ choose } y = -x \text{ Then } \exists y \in \mathbb{R} \text{ and } x + y = x + (-x) = 0)$$

04. There exists real number x for all real number y such that $xy = 0$

$$\exists x \in \mathbb{R} \forall y \in \mathbb{R}, xy = 0 \text{ is F}$$

Consider the following open propositions with universe $\mathbb{R}$.

a. $R(x): x^2 \geq 0$.

b. $P(x): (x+2)(x-3) = 0$.

c. $Q(x): x^2 < 0$.

" $\forall x \in \mathbb{R}, x^2 \geq 0$ " is T

" $\exists x \in \mathbb{R}, (x+2)(x-3) = 0$ " is T

" $\forall x \in \mathbb{R}, x^2 < 0$ " is F

Then $R(x)$ is always true for each $x \in \mathbb{R}$,

$P(x)$ is true only for $x = -2$ and $x = 3$,

$Q(x)$ is always false for all values of $x \in \mathbb{R}$.

Hence, given an open proposition $P(x)$, with universe U , we observe that there are three possibilities.

a. $P(x)$ is true for all $x \in U$.

b. $P(x)$ is true for some $x \in U$.

c. $P(x)$ is false for all $x \in U$.

Write the following statements using quantifiers.

i. For each real number $x > 0$, $x^2 + x - 6 = 0$.

ii. There is a real number $x > 0$ such that $x^2 + x - 6 = 0$.

iii. The square of any real number is nonnegative. (iii) " $\forall x \in \mathbb{R}, x^2 \geq 0$ " is T

Ans: i. " $\forall x \in \mathbb{R}, x^2 + x - 6 = 0$ " is F
ii. " $\exists x \in \mathbb{R}^+, x^2 + x - 6 = 0$ " is T (Since when $x=2$, $2^2+2-6=0$)

Question 1: Let $P(x)$ be the predicate " $x^2 - 1 = 0$ " where x is a real number.

Determine the truth value of $\exists x P(x)$.

$\exists x P(x)$ is T (Since $P(1): 1^2-1=0$ is True)

Question 2: Let $Q(x, y)$ be the predicate " $x < y$ " where x and y are integers. What does $\forall x \exists y Q(x, y)$ mean in words?

For each integer x there exists integer y such that $x < y$. is T

Question 3: Let $R(x)$ be the predicate " x is even" where x is an integer. Write the statement "All integers are even" using predicate logic.

" $\forall x \in \mathbb{Z}, R(x)$ "

$x \in \mathbb{Z}$

$R(x): x$ is even

Question 4: Let $S(x)$ be the predicate " x is a mammal" and $T(x)$ be " x can fly" where x is an animal. How would you express "Some mammals can fly" using predicate logic?

$S(x): x$ is a mammal
 $T(x): x$ can fly

$\exists x, S(x) \Rightarrow T(x)$

Question 5: Let $P(x, y)$ be the predicate " x is the parent of y " where x and y are people. Express "Everyone has a parent" using predicate logic.

$\forall x \exists y, P(x, y)$

$P(x, y): x$ is the parent of y

Question 6: Let $Q(x)$ be the predicate " x is prime" where x is a positive integer. What is the negation of $\forall x Q(x)$?

$\forall x \in \mathbb{N}, Q(x)$

"For all positive integer x is prime"

Ans: x is prime, $x \in \mathbb{N}$

There exists positive integer x such that x is not Prime

Question 7: Let $R(x, y)$ be the predicate " $x + y = 10$ " where x and y are integers. Is the statement $\forall x \forall y R(x, y)$ true or false? Explain why.

Question 8: Let $S(x)$ be the predicate " x is a multiple of 5" where x is an integer. Express the statement "There is no integer that is a multiple of 5" using predicate logic.

Question 9: Let $T(x, y)$ be the predicate " x is taller than y " where x and y are people. Express the statement "Nobody is taller than themselves" using predicate logic.

Question 10: Let $R(x)$ be the predicate " x is a vowel" where x is a letter in the English alphabet. Write the following statement using predicate logic: "There is a vowel in the English alphabet."

Note:

(1) The Universal quantifier " $\forall$ "

Let $p(x)$ be a predicate, where x in D .

Suppose 'For any x in non-empty subset S of D , $p(x)$ ' is true.

Then, 'For any x in non-empty subset S of D , $p(x)$ ' is a proposition.

We write this proposition as ' $\forall x \in S, p(x)$.'

Which means, 'For any x in non-empty subset S of D , $p(x)$ ' is true.

i.e., $\forall x \in S, p(x)$.

Example:

Let $p(x)$: x is even, where x in $D = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, \}$.

Suppose $S = \{2, 4, 6, 8, 10\}$.

For any x in non-empty subset S of D , $p(x)$ is true.

i.e., $\forall x \in S, p(x)$.

Notation: $\forall x P(x)$, where $\forall$ is called the universal quantifier.

Read as "for all x $P(x)$ " or "for every x $P(x)$ ".

An element x for which $P(x)$ is false is called a counterexample of $\forall x P(x)$.

Other ways to say $\forall$:

- All of...
- For each....
- Given any.....
- For any....
- For arbitrary....

For example,

Let $P(x): x^2 > x$, where x is a real number.

Then " $\forall x P(x)$ " in English "for all $x, x^2 > x$ " or we can say "for any $x, x^2 > x$ " or in other words "given any $x, x^2 > x$ "

An element $x = \frac{1}{2}, \frac{1}{4} > \frac{1}{2}$ is false. i.e. $P\left(\frac{1}{2}\right)$ is false.

So, $\frac{1}{2^2} > \frac{1}{2}$ is a counter example for "for all $x, x^2 > x$ ", where x is a real number.

Example 1:

Consider the predicate $|x| \geq 0$, where $x \in \mathbb{R}$.

Let $P(x): |x| \geq 0$.

Then, $\forall x \in \mathbb{R}, P(x)$ is true;

i.e. $\forall x \in \mathbb{R}, |x| \geq 0$.

Example 2:

Consider the predicate $x > \frac{1}{2}$, where $x \in \mathbb{R}$

Let $P(x): x > \frac{1}{2}$.

Then $\forall x \in \mathbb{R}, P(x)$, is false, since for instance $0 \in \mathbb{R}$, but $P(0)$ is false.

i.e. not $(\forall x \in \mathbb{R}, P(x))$.

However $\forall x \in \mathbb{N}, P(x)$ is true.

i.e. $\forall x \in \mathbb{N}, P(x)$.

(2) The Existential quantifier " $\exists$ "

Let $p(x)$ be a predicate, where x in D .

Suppose 'There is x_0 in non-empty subset S of D , $p(x_0)$ ' is true.

Then, 'There is x_0 in non-empty subset S of D , $p(x_0)$ ' is a proposition.

We write this proposition as ' $\exists x_0 \in S, p(x_0)$ '.

Which means, 'There is x_0 in S such that $p(x_0)$ ' is true.

i.e., $\exists x_0 \in S, p(x_0)$.

Notation: $\exists x P(x)$, where $\exists$ is called the **existential quantifier**.

Other ways to read $\exists$:

"There is an x such that $P(x)$."

"There is at least one x such that $P(x)$."

"For some x , $P(x)$."

"We can find an x , such that $p(x)$ "

Examples:

(i). Consider the predicate $2x > 7$ where $x \in \mathbb{N}$

Let $P(x) : 2x > 7$

When $x = 5$, $x \in \mathbb{N}$ and $2x > 7$ is true (since $10 > 7$)

i.e. when $x = 5$, $x \in \mathbb{N}$ and $P(5)$ is true

$\therefore \exists x \in \mathbb{N}$, $P(x)$ is true

i.e. $\exists x \in \mathbb{N}$, $2x > 7$.

(ii). Consider the predicate $x^2 > 1$, where $x \in \mathbb{R}$

Let $A = \{0, 1, -1\}$ and $P(x) : x^2 > 1$

$P(0), P(1), P(-1)$ are false.

$\therefore \exists x \in A$, $P(x)$ is false.

i.e. not $(\exists x \in A, P(x))$.

However $\therefore \exists x \in \mathbb{N}$, $P(x)$ (since when $x = 2$, $x \in \mathbb{N}$ and $P(2)$ is true)

i.e. $\exists x \in \mathbb{N}$, $x^2 > 1$.

End of D5